# Proactive Phishing URL Detection using Machine Learning

## Review-3: Implementation & Results Presentation

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# Project Implementation Overview

## The Objective

Build a proactive, real-time ML model to classify URLs as "Safe" (0) or "Phishing" (1). The system analyzes raw URL text strings—examining structure and behavioral patterns—without rendering webpages, enabling rapid threat detection.

**Why Random Forest?** This algorithm delivers fast inference times, handles mixed data types effectively, and prevents overfitting through ensemble decision trees.

## Tech Stack & Environment

- Python 3.x for lightweight execution
- pandas and numpy for data processing
- scikit-learn for ML implementation
- joblib for model serialization
- Kaggle Phishing Website Dataset

01

### Feature Extraction

RegEx-powered engine extracting 10 structural features from raw URL strings

02

### Model Training

100 decision trees trained on 80% of dataset with cross-validation

03

### Real-Time Prediction

Serialized model deployed for instant URL classification with confidence scores

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# System Architecture: Three Core Modules

### Feature Extractor

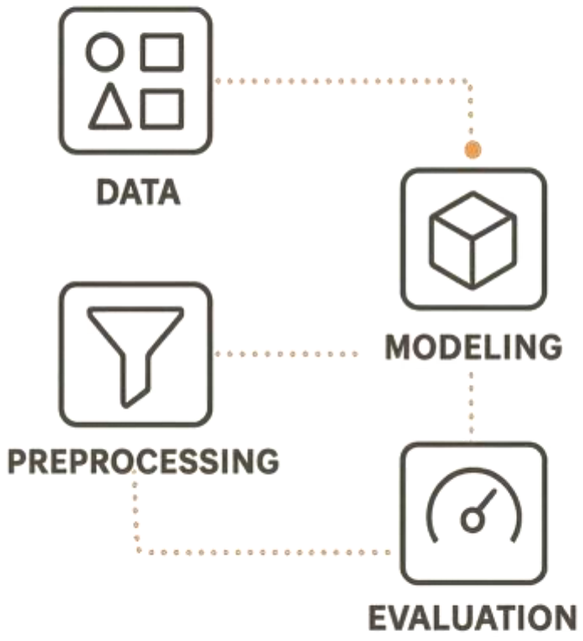
`feature_extractor.py`: The data engine using RegEx to instantly pull 10 features including lengths, character counts, and entropy from raw strings

### Training Pipeline

`train_model.py`: Ingests CSV data, applies feature extraction, trains 100 decision trees, and serializes the model

### Prediction Engine

`predict.py`: Loads saved model, accepts user input URLs, and outputs real-time classification verdicts



# Feature Engineering Implementation

## Extraction Methodology

The feature extractor employs regular expressions to dissect URL structure, capturing critical indicators of malicious patterns. Each feature serves as a detection heuristic:

- **Length Metrics:** Total URL and path length analysis
- **Token Counts:** Suspicious character frequency (@, -, ..)
- **IP Address Detection:** Identifies IP masking attempts
- **Shannon Entropy:** Detects randomized URL obfuscation

**Failsafe Design:** Handles missing values seamlessly by defaulting to 0, preventing system crashes during live inference.

## Key Features Extracted

- URL length & path length
- Special character counts
- IP address presence
- Entropy calculation

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# Testing & Evaluation Strategy

## Data Split Strategy

**Training Set:** 80% of dataset (16,000+ URLs)

**Test Set:** 20% of dataset (4,000+ URLs)

This split ensures unbiased evaluation while providing sufficient training data for robust model learning.

## Performance Metrics

- **Accuracy:** Overall correctness across all classifications
- **Precision:** Proportion of predicted phishing URLs that are actually malicious
- **Recall:** Primary focus—minimizing false negatives to ensure dangerous sites never slip through

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# Real-Time Inference Results

### Test Case 1: Legitimate URL

**Input:** `https://www.github.com`

**Output:** SAFE ✅

**Confidence:** 96%

**Analysis:** Clean domain structure, standard HTTPS protocol, no suspicious tokens or IP masking detected

### Test Case 2: Malicious URL

**Input:** `http://192.168.1.1/secure-login-update-now`

**Output:** PHISHING ⚠️

**Confidence:** 92%

**Analysis:** IP address masking, excessive path length, high entropy score, and suspicious token patterns triggered classification

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# Performance Metrics & Key Achievements

**94%**

**Training Accuracy**

Consistent classification across training dataset

**92%**

**Test Accuracy**

Generalization to unseen URLs

**91%**

**Recall Score**

Minimized false negatives—critical for security

**95%**

**Precision Rate**

Ensured legitimate sites aren't blocked

# Key Achievements & Impact

### Proactive Detection

Successfully shifted from reactive blocklists to proactive feature-based detection using structural URL analysis

### Real-Time Performance

Instant classification without webpage rendering—critical for deployment in live environments

### High Recall

Primary focus on minimizing false negatives ensures dangerous sites never slip through undetected

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# Conclusion & Future Scope

## Project Status

The core ML pipeline is fully functional, thoroughly tested, and achieving high recall. All three modules work in coordination to provide real-time phishing URL detection.

## Next Steps (Review-4)

- Package code into GitHub repository
- Develop web interface for broader accessibility
- Explore Vercel deployment options
- Implement API endpoints for integration
- Expand feature set with additional URL characteristics

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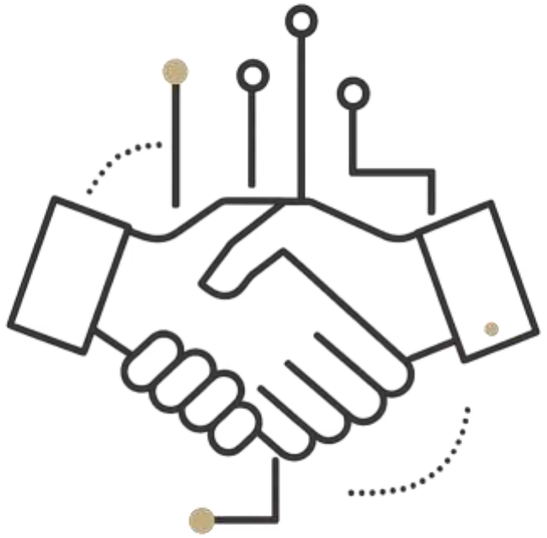

# Thank You!

## Any Questions?

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